YUKUN ZHOU

◇ Hong Kong ◇ yukunzhou450@gmail.com ◇ personal web

EDUCATION

City University of Hong Kong (CityU of HK)
Bachelor of Computer Science, Minor in Creative Media, 2019 - 2023

- CGPA: 3.8/ 4.30 (Top 14 out of 144 CS students), average 3.95 for all major courses
- First Class Honor

University of Southern California (USC)
Master of Science, Computer Science - Game Development 2023 - 2025

- CGPA: 3.87/ 4.0

RESEARCH OUTPUT

Manuscripts accepted with minor revisions, ACM Transactions on Graphics (TOG) 2026
Anonymous author, Yukun Zhou, Other Authors. About the enhancement of Vertex Block Descent
Manuscript under Review
Yukun Zhou*, Other Authors. A paper on games, ICT4D, player experience, and well-being.

RESEARCH EXPERIENCE

Research Assistant
Advised by: Dr Chenfanfu Jiang June 2026- Current
Univeristy of California, Los Angeles *Los Angeles*

- Developing real-time rod simulation with contact and real robot perception and control policies for the complicated cable manipulation task.

Joint RA–PhD Program - Physics-Based Simulation
Advised by: Dr Zhongkai Zhang, Dr Hongbin Liu July 2025- June 2026
Chinese Academy of Sciences (Hong Kong Institute of Science and Innovation) and Chinese University of Hong Kong *Hong Kong*

- **First-author Manuscript in Preparation**. Finished main experiments and producing a paper draft, about accurate rod-insertion simulation algorithms based on Projective Dynamics, Cosserat rods using Lagrangian constraint-based methods, and modelling thickness-aware rod contact pressure and rotational frictions.
- Discussed the algorithms and worked on the code and experiments of the Vertex Block Descent enhancemnet project and paper.
- Developed and reproduced sdf softbody contact, DiSect differentiable cuttings, FEM, Projective Dynamics cutting with Low-rank update methods, lung puncturing by Warp.
- Completed human brain 3D reconstruction from MRI images using neural networks.

Summer Research Intern - Fluid Simulation about Neural and Particle Flow Map
Advised by: Dr Bo Zhu June 2024-Sep 2024
Georgia Institute of Technology *Atlanta, USA*

- Replicated Neural Flow Map algorithm by Taichi and develop the real-time 3D stable fluid simulation by OpenGL Compute Shader.

- Developed the algorithm acceleration with Long-range-mapping-short-range-projection approaches on Particle Flow Map method with 80% speed improvement.

Part-time Research Assistant - HCI Research on the Game and Players
Advised by: Dr Zhicong Lu June 2022-Sep 2023
City University of Hong Kong *Hong Kong*

- Produced a paper **under review** as the only first author.
- Completed the research project that contributes to games and ICT4D in HCI and provides insights to enhance their well-being by suggesting design implications and further application scenarios.

WORK EXPERIENCE

Grader of Physical-based Simulation and Animation Course at USC Work as the grader for the course by Dr. Jernej Barbic Jan 2025-May 2025

Data Science Intern (Full-time, Four Days A Week) Siemens - Mobility, Hong Kong June 2021-March 2022

- Designed and developed map matching, route matching and traffic flow prediction algorithms and machine learning time series models (bi-LSTM, Encoder-decoder LSTM, ARIMA).
- Database manipulate and data process by Python to support database building and maintaining.

RELATED PROJECTS

Physics-based Simulations Implemented Mass Spring system, Keyframe insertion with Quaternion and Euler approach, Inverse Kinematic with Pseudo Inverse and Tikhonov regularization method in C++.

Game Development: Lucidmare, PL-23 and Donuts! by Unity Developed Unity 3D programs in C, lead and design in three games: Lucidmare(Demo), Omnis(Demo), Donuts! (Published on Steam)

3D Human Rendering A 3D graphic course group project, achieving URP, hair anisotropy, and skin separable subsurface scattering under Unity engine. Learn and deploy skin separable subsurface scattering in glsl.

Hand-scratching Gesture Recognition Android Application for AR Glasses Advised by: Dr Kening Zhu, *City University of Hong Kong.* Developed an Android application with CNN models to analyze the sound of scratching the AR glasses leg in different ways in real-time to take the recording and real-time recognition function, and control the AR glasses.

NLP for Finance Advised by: Dr Linqi Song, *City University of Hong Kong.* Applied deep learning knowledge and developed the BERT model to analyze the sentiment of large amounts of comments in PyTorch.

AWARDS & HONORS

- Entrance Full Tuition Scholarships 2019-2023
- Dean's list awards for outstanding academic performance 2019-2023

SKILLS

Other Technical Skills	AI programming, Drone Programming, Digital Art and Installation, Game Design and Development
Programming Language	Python, C, Java, C++, Processing, JavaScript, HTML, CSS, SQL
Language	Native Mandarin, Fluent English, Basic Cantonese